

BE 505: Computational Biology

Course Code	BE 505
Course Name	Computational Biology
L-T-P-C	2-0-2-3
Prerequisites	IC136, BE304, or consent of faculty
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021)

Course Contents

- **Pattern matching**
inding regulatory sequences in DNA, Motif discovery, Exhaustive search, Greedy motif clustering, wordlets and motifs refinements, Probabilistic solutions (expectation maximisation, Gibbs sampling). [2 Lectures]
- **Genomic Analysis**
 - a. Genome assembly with Graphs and Networks,
 - b. Gene prediction, Hidden Markov models, Viterbi, expectation maximisation)
 - c. Next Generation Sequencing analysis (short read mapping, ChIP-seq and RNA-seq analysis)
 - d. Comparative genomics and genome rearrangements
 - e. Population genomics, Medical genomics, Personal genomics, disease epigenomics, Systems approaches to disease. [14 Lectures]
- **Phylogenetic Inferences**
Introduction to phylogenetic inferences, Alignments to Distances, probabilistic models of divergence (Jukes Cantor, Kimura, hierarchy), Distances to trees, types of trees, algorithms for tree building (UPGMA, neighbor joining), optimality (least squared error, minimum evolution), Alignments to trees, alignment scoring given a tree, parsimony, greedy vs dynamic programming, maximum likelihood, Max-a-Posteriori, bootstrapping, Tree visualisation. [3 Lectures]
- **Structure prediction and Molecular modelling**
RNA and protein structure prediction methods, homology modelling and ab – initio structure prediction methods, models of proteins, discrete conformational search, binding and docking, molecular dynamics simulations. [5 Lectures]
- **Biological network analysis and modelling** Gene regulatory networks, Protein interaction networks, Logic modelling of Cell signalling networks, network modelling, formulating models, nonlinear dynamics and stability, steady-state problems, parameter fitting and estimation, basic overview of the modeling of metabolic networks in genome scale by Flux Balance or modeling of reaction kinetic for smaller networks/pathways [4 Lectures]

Laboratory Modules

The below mentioned 10 topics will be covered over the 14 weeks:

- Pattern matching
- Genome assembly
- Gene prediction (prokaryotic and eukaryotic)
- Read mapping and NGS data-analysis
- Comparative genomics methods
- Population genomics methods
- Phylogenetic and analysis molecular evolution
- Protein structural analysis (Secondary structure prediction and homology modelling)
- Protein structural analysis (Ab-initio molecular modelling and simulation, docking)
- Biological networks analysis

Text Books

1. Peter Clote, **Computational Molecular Biology: An Introduction** in Biomaterials Science, 2000.
2. Neil C. Jones and Pavel A Pevzner, **An Introduction to Bioinformatics Algorithms (Computational Molecular Biology)**, 2004.

References

1. Ion Mandoiu (Author), Alexander Zelikovsky, **Bioinformatics Algorithms: Techniques and Applications: 03**, Wiley Series in Bioinformatics, 2008.
2. Arthur M. Lesk, **Introduction to Bioinformatics**, 3rd edition, Oxford University Press.
3. Richard Durbin, Sean R. Eddy, Anders Krogh, Graeme Mitchison, **Biological Sequence Analysis-Probabilistic Models of Proteins and Nucleic Acids**, Cambridge University Press.
4. Jonathan Pevsner, **Bioinformatics and Functional Genomics**, 3rd edition, Wiley-Blackwell.